



数字逻辑与微处理器系统设计

实验1: Introduction to Development Tools

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实验课概述



实验课环节共32学时，分8次课完成，规划如下：

第一次课：掌握FPGA行为级仿真能力，完成加法器测试仿真；

第二次课：掌握FPGA上板调试能力，完成基于状态机的倒计时器上板；

第三次课：实验1&实验2分组验收；

第四、五次课： ALU设计、实现、优化，以及验收；

第六、七、八次课： MCU设计、实现、优化，以及验收；



目录



1. 课程设计介绍
2. FPGA开发板简介
3. Vivado开发工具简介
4. 全加器仿真例程
5. VHDL编程语言学习资源
- 6. 任务: 8-bits加法器(仿真)**



- 课程设计1：ALU的设计与实现
- 课程设计2：MCU及其外围电路的设计与实现
- 提交内容：设计报告、源代码、测试程序及测试结果。抄袭将取消考试资格。
- 考核方式：竞赛式考核 + 应用设计（排序、数字信号处理）
- 3人一组组队完成，设组长1人，组员2人。教学团队给出设计的平均成绩，由组长分配每个成员课程设计成绩；如2名团队成员对组长分配有异议，可向教学团队提出复议，根据实际情况重新分配。
- 考核时间结课后一周，报告提交时间另行通知。

竞赛式考核

- 基于所设计MCU完成两个应用，分别按照速度和效率完成排名
- 每个榜单第一名96分，第二名92分，以此类推。
- 根据提交课程设计报告质量 ± 4 分。
- 每个榜单第一名从其余榜单删除，屠榜队从所有榜单删除。

赛题1		赛题2	
速度榜单	效率榜单	速度榜单	效率榜单
第一名	第一名	第一名	第一名
第二名	第二名	第二名	第二名
.....			

考核注意事项

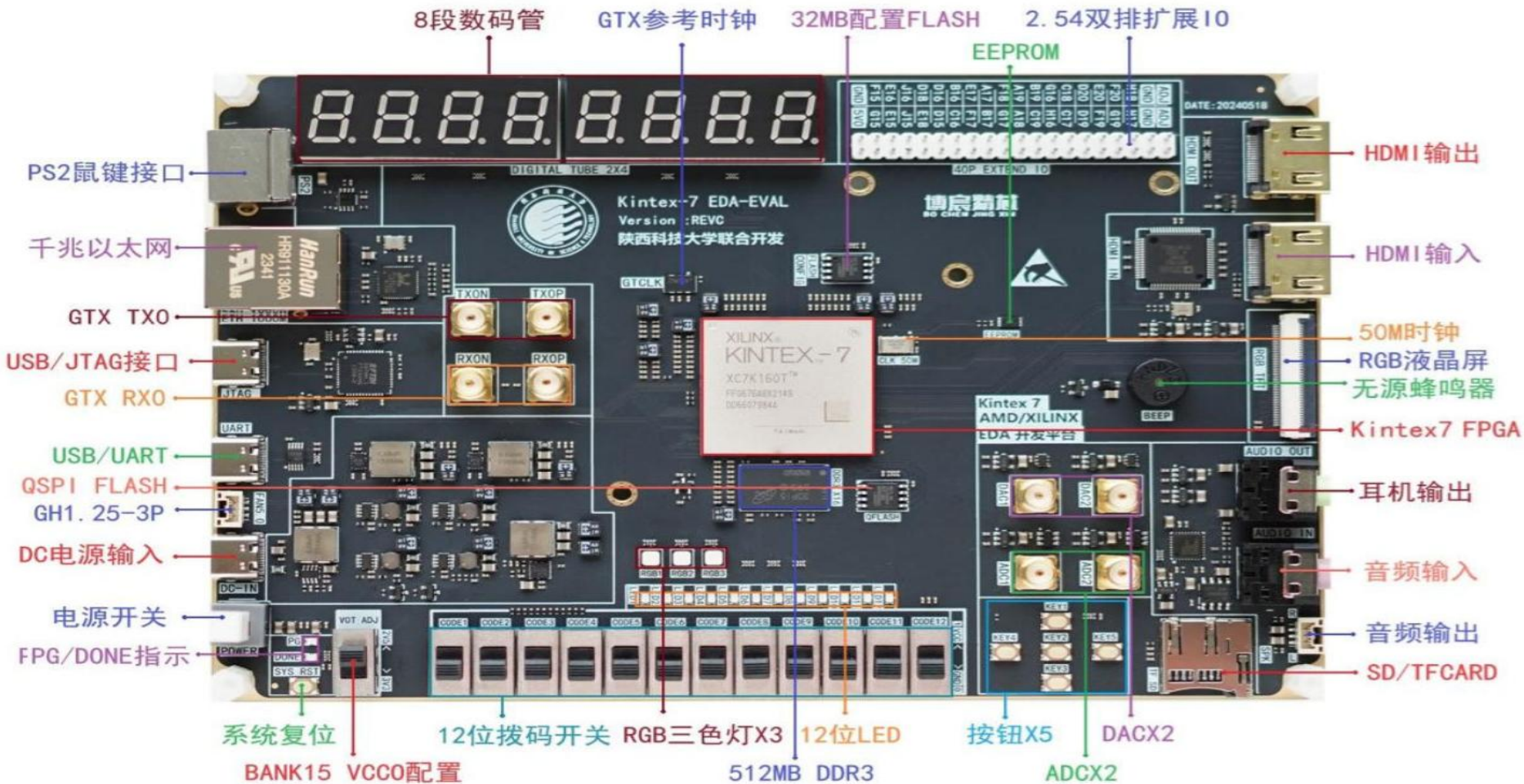
- 处理速度统计方法：完成所有计算结果的**时钟频率/时钟周期数**，其中时钟频率取时序报告无违例频率与上板时钟频率的较小值；
- 处理效率统计方法：**处理速度/硬件开销**，其中硬件开销折算方法为 $6 * \text{LUT} + 10 * \text{FF}$ ；
- MCU只能执行ARM指令集范围内的指令，不允许额外扩展指令；
- MCU微架构不限，流水、并行、单指令多操作等任意优化均可使用；
- 教学团队会提供板级Testbench，也可以自行搭建测试环境；
- 加分项：采用创新技术优化MCU性能指标，均可作为加分项，分值由教学团队共同评定

FPGA开发板简介

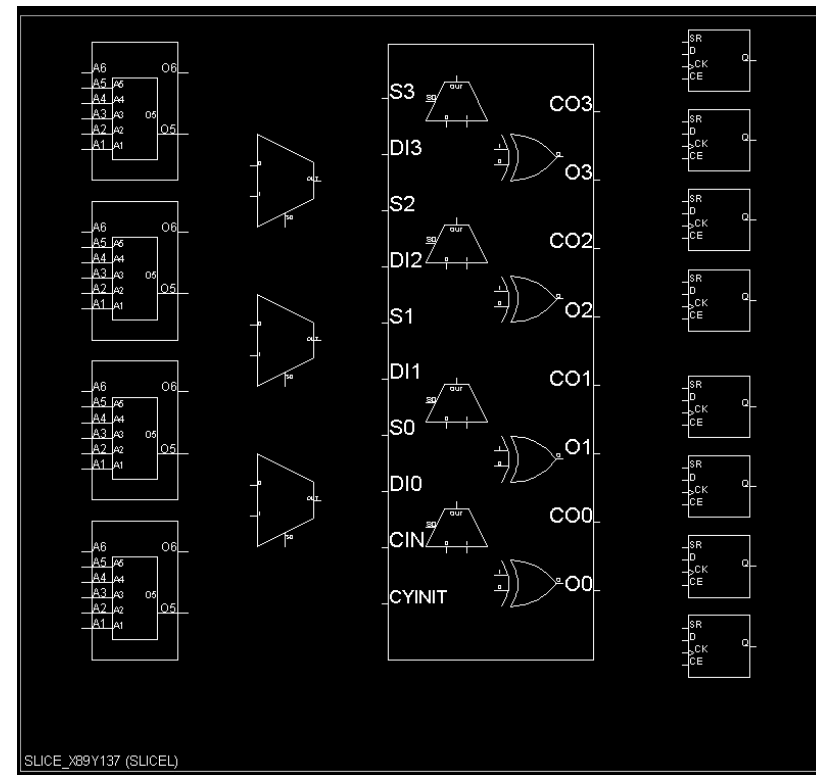
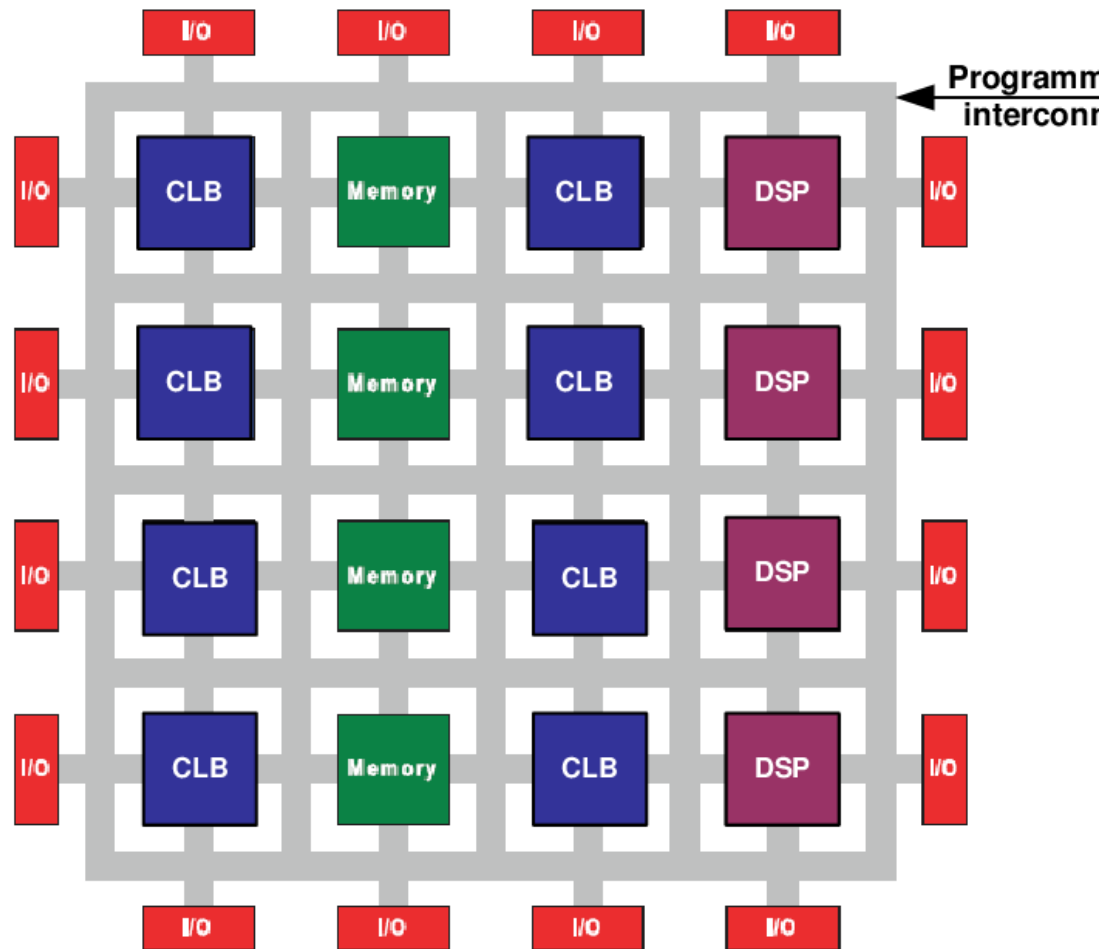


Xilinx Kintex®-7 FPGA 芯片 XC7K160T-2FFG676-I

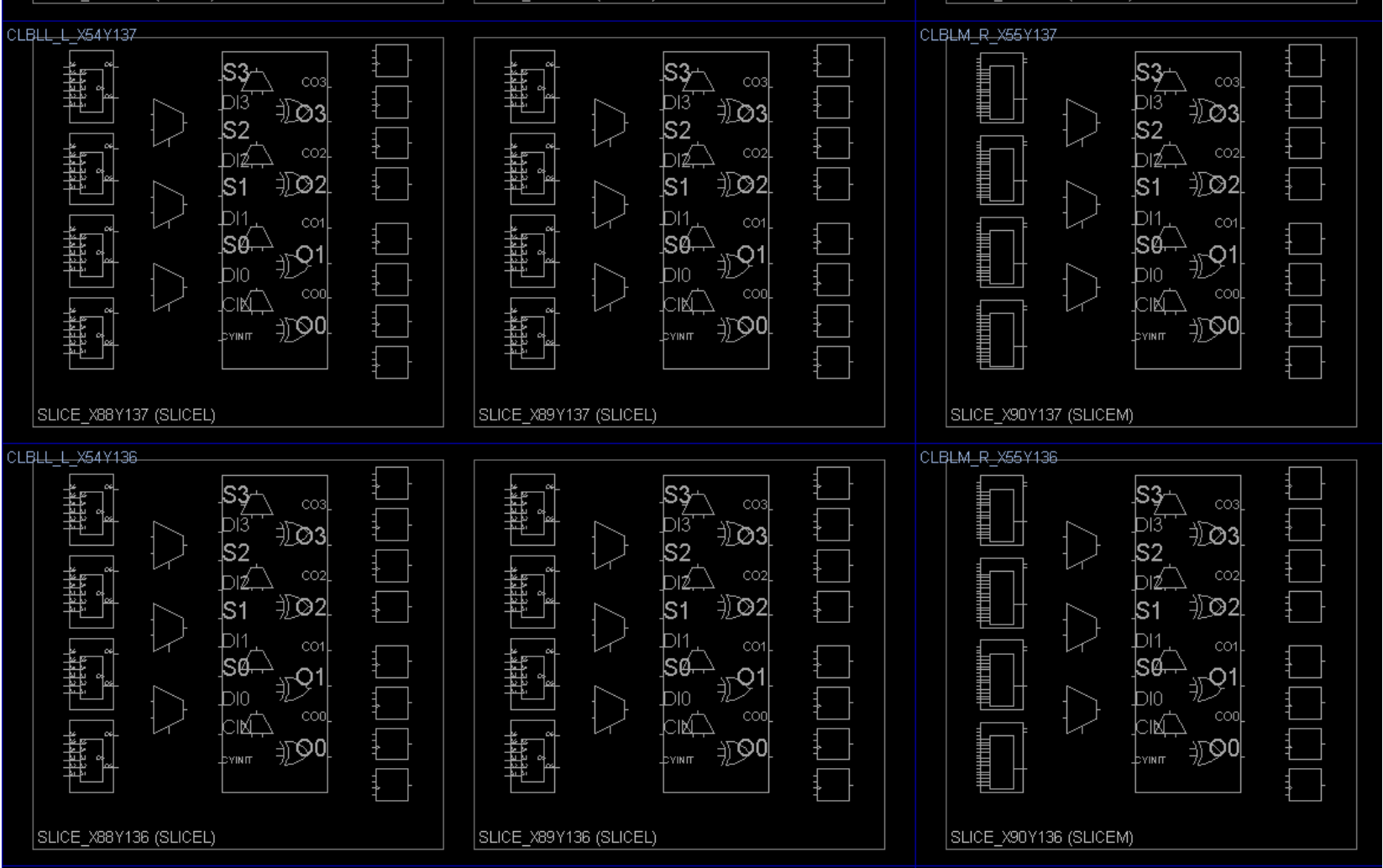
K7EDA-EVAL开发平台功能标注



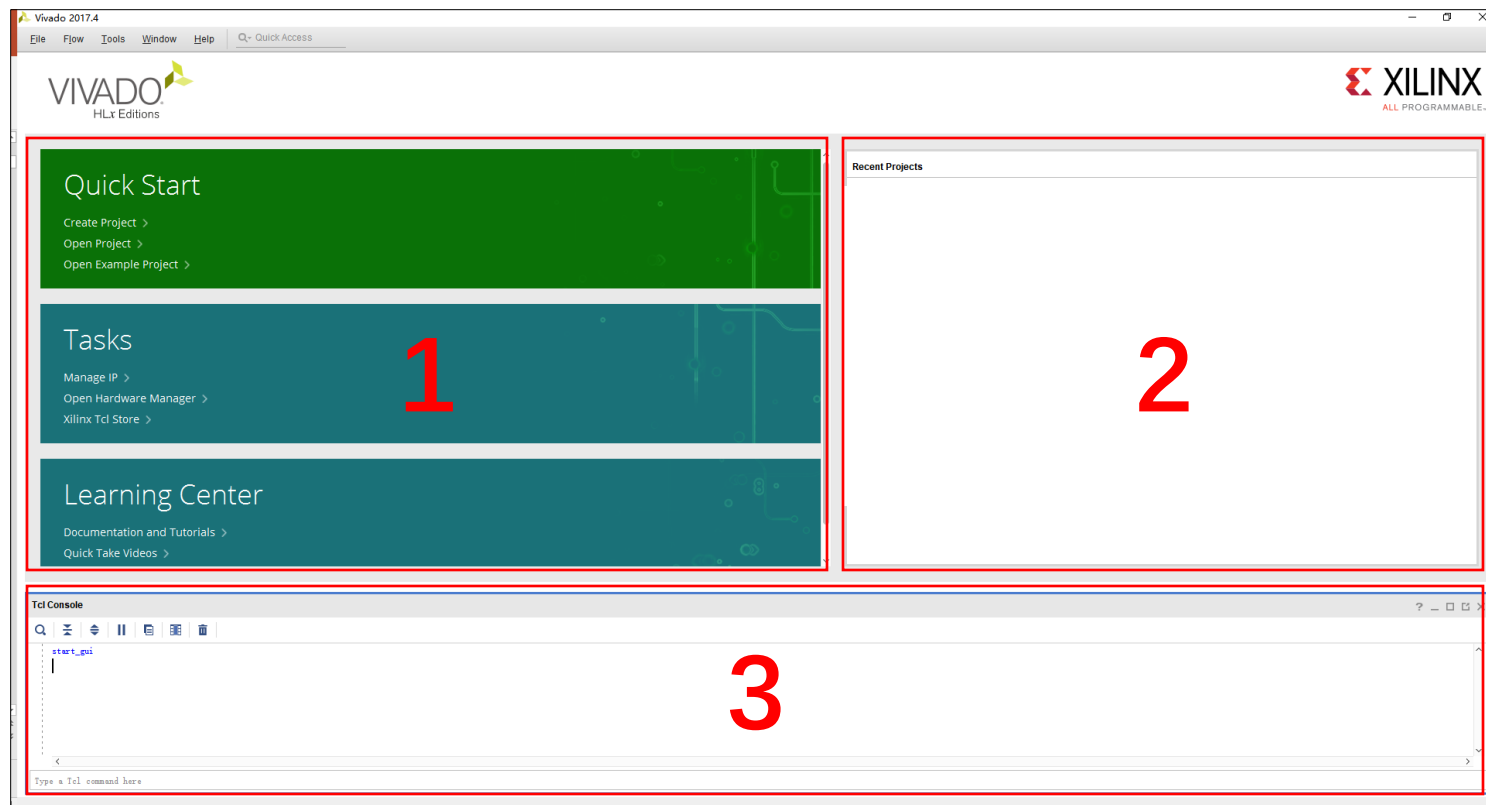
FPGA开发板简介



FPGA开发板简介



Vivado开发工具简介

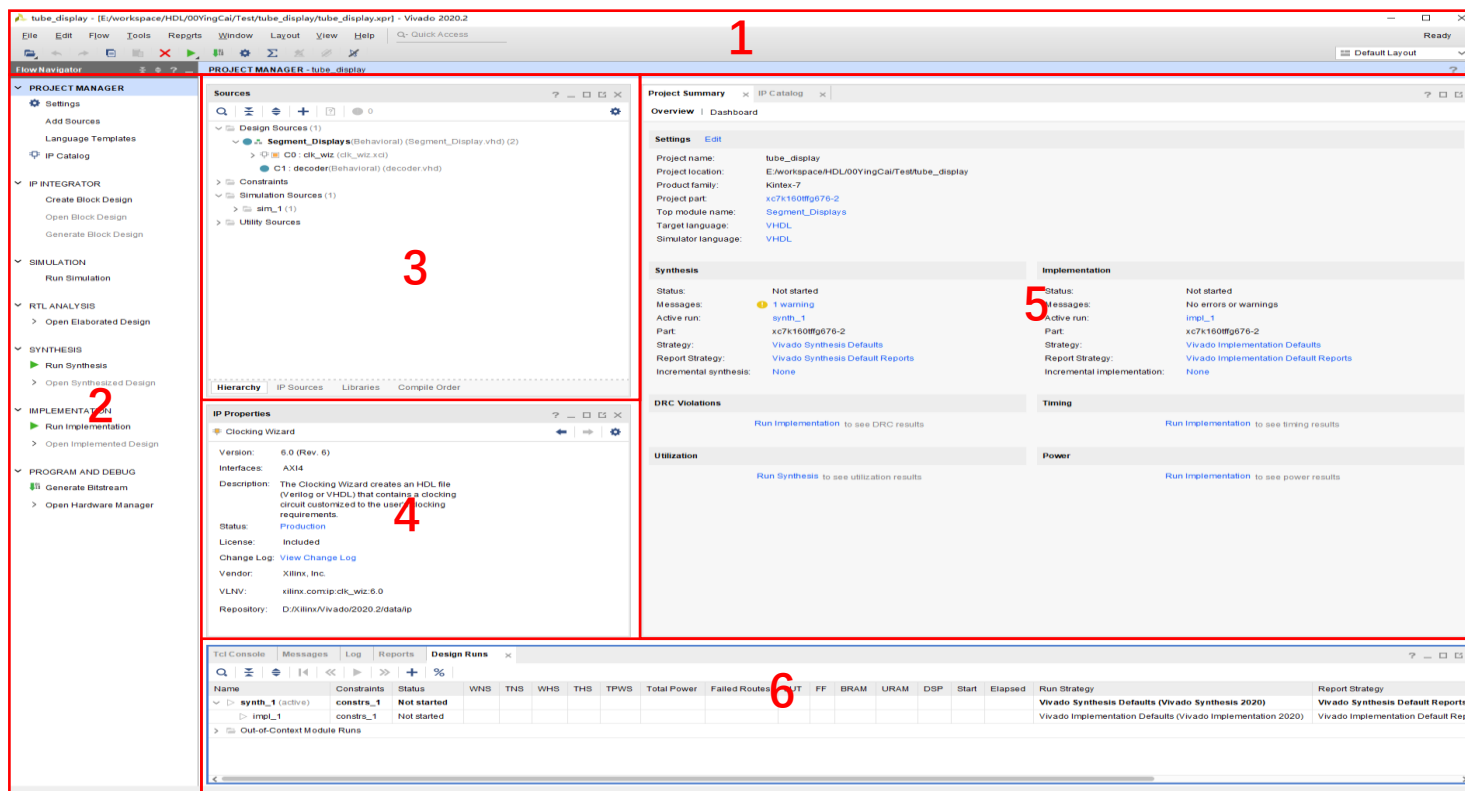


1区域中常用的功能有：Quick Start中的Create Project和Open Project以及Tasks中的Open Hardware Manager

2区域中是最近打开过的工程

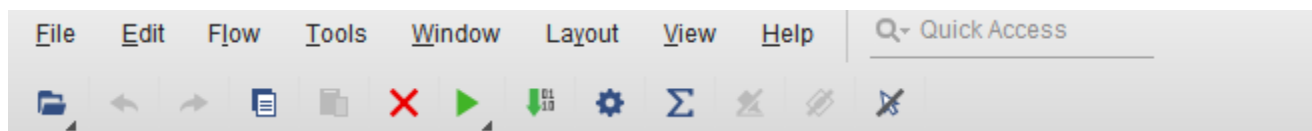
3区域是使用Tcl命令的操作区域

Vivado开发工具简介



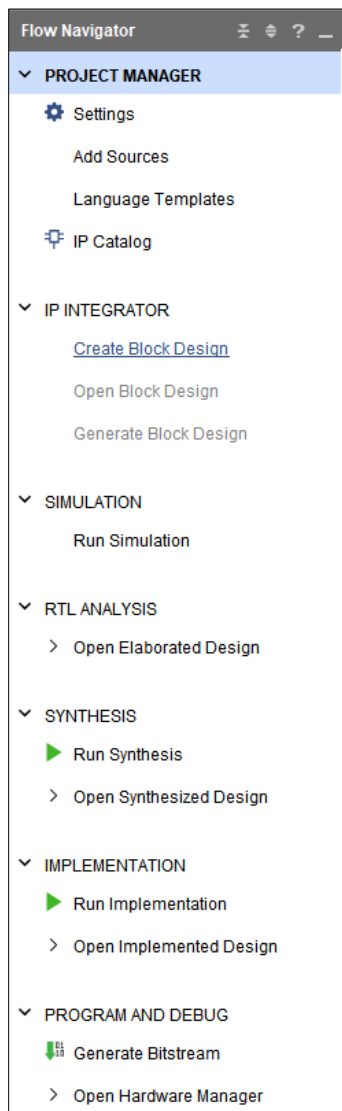
Vivado的主界面分为以下的6大部分，下面做介绍。

Vivado开发工具简介



工具栏：大部分常用功能在其它区域都有按钮，使用起来更加方便。工具栏内容较多，大家感兴趣可以去享受下探索的乐趣。

Vivado开发工具简介



项目管理：工程设置，新建或添加源文件（约束、代码、仿真源文件），语法模板、IP核生成

Block Design：图形化设计

仿真

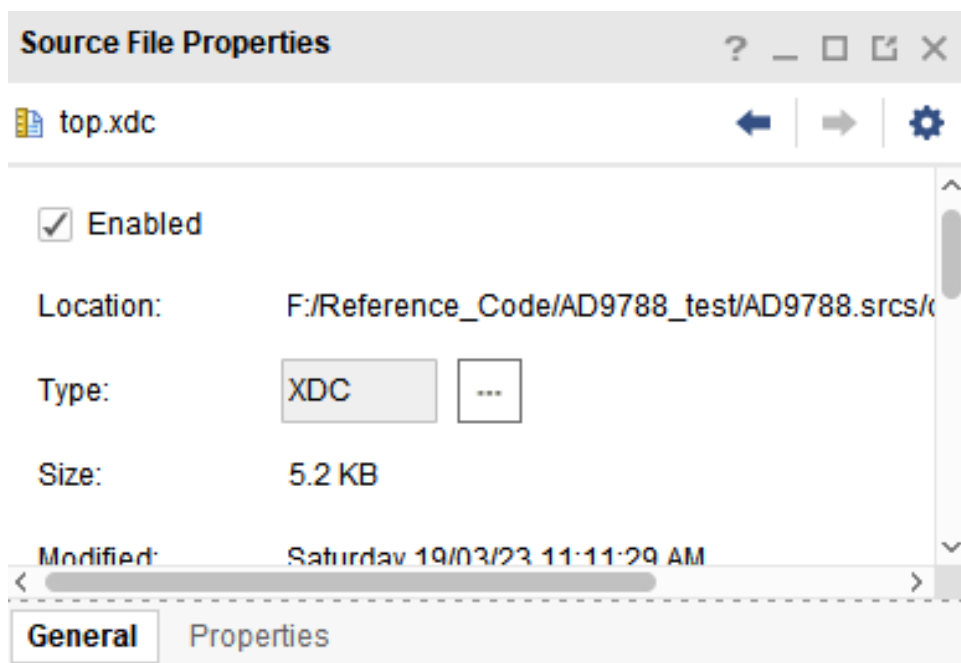
RTL分析：DRC报告，电路原理图

综合：代码综合，时序约束，Set Up Debug，生成门级网表，各种报告

实现：优化，根据网表做映射，布局布线

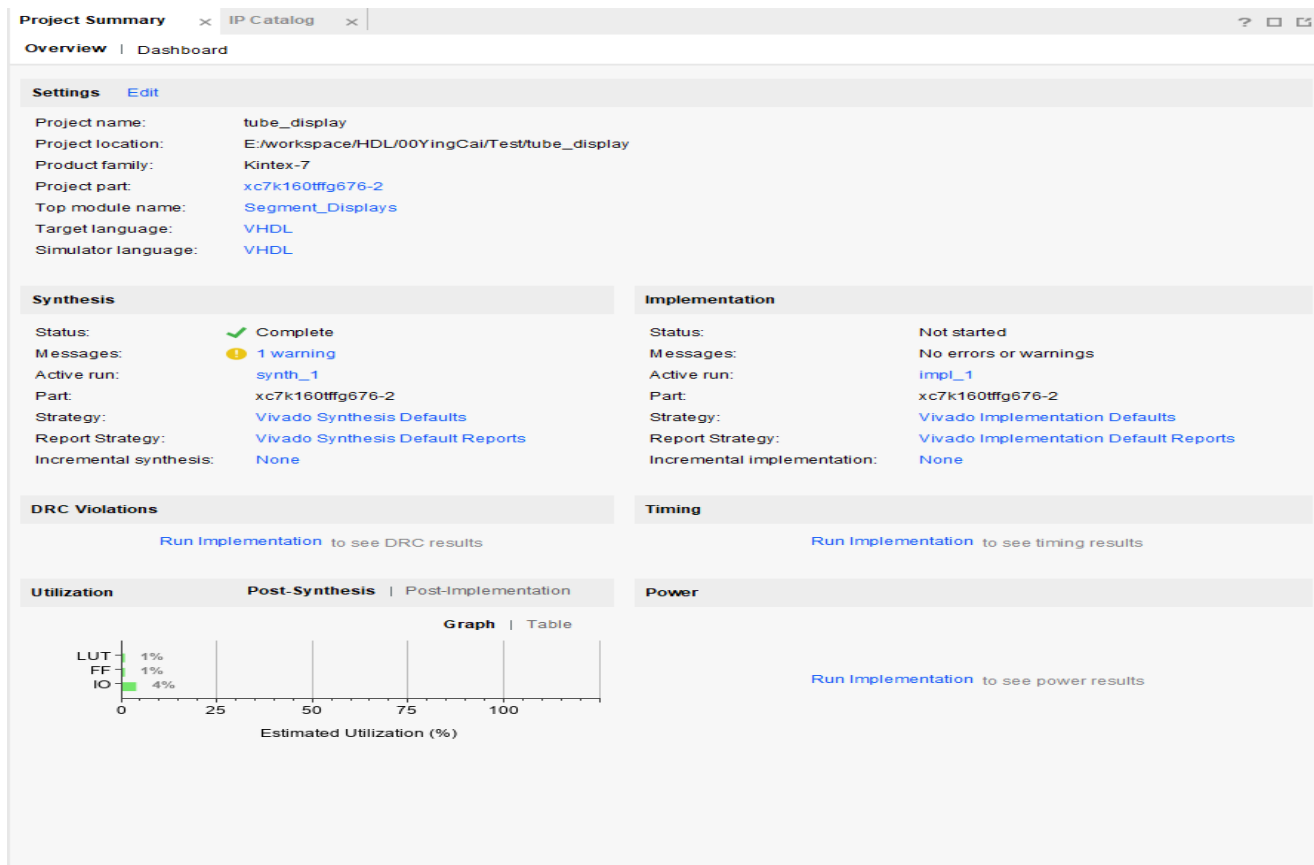
生成比特流以及下载

Vivado开发工具简介



主要显示选中文件的属性。例如，图中显示的是top.xdc约束文件的位置、类型、大小、修改日期等等。可以关掉。

Vivado开发工具简介



工程设置、综合结果、实现结果、DRC、时序、功耗、资源消耗等等。

Vivado开发工具简介

A screenshot of the Vivado Tcl Console window. The window has a title bar with tabs for 'Tcl Console', 'Messages', 'Log', 'Reports', and 'Design Runs'. Below the tabs is a toolbar with icons for search, zoom, run, pause, copy, paste, and delete. The main text area displays the following output:

```
Finished scanning sources
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1704] No user IP repositories specified
INFO: [IP_Flow 19-2313] Loaded Vivado IP repository 'H:/vivado2017.4/Vivado/2017.4/data/ip'.
open_project: Time (s): cpu = 00:00:12 ; elapsed = 00:00:06 . Memory (MB): peak = 844.512 ; gain = 135.949
update_compile_order -fileset sources_1
```

At the bottom, there is a text input field with the placeholder text 'Type a Tcl command here'.

Tcl命令窗口、综合信息如警告错误等、报告、一些运行信息

Vivado开发工具简介



第一步： 打开 Vivado， 稍等软件启动后， 进入如下界面



Quick Start

[Create Project >](#)
[Open Project >](#)
[Open Example Project >](#)

Tasks

[Manage IP >](#)
[Open Hardware Manager >](#)
[XHub Stores >](#)

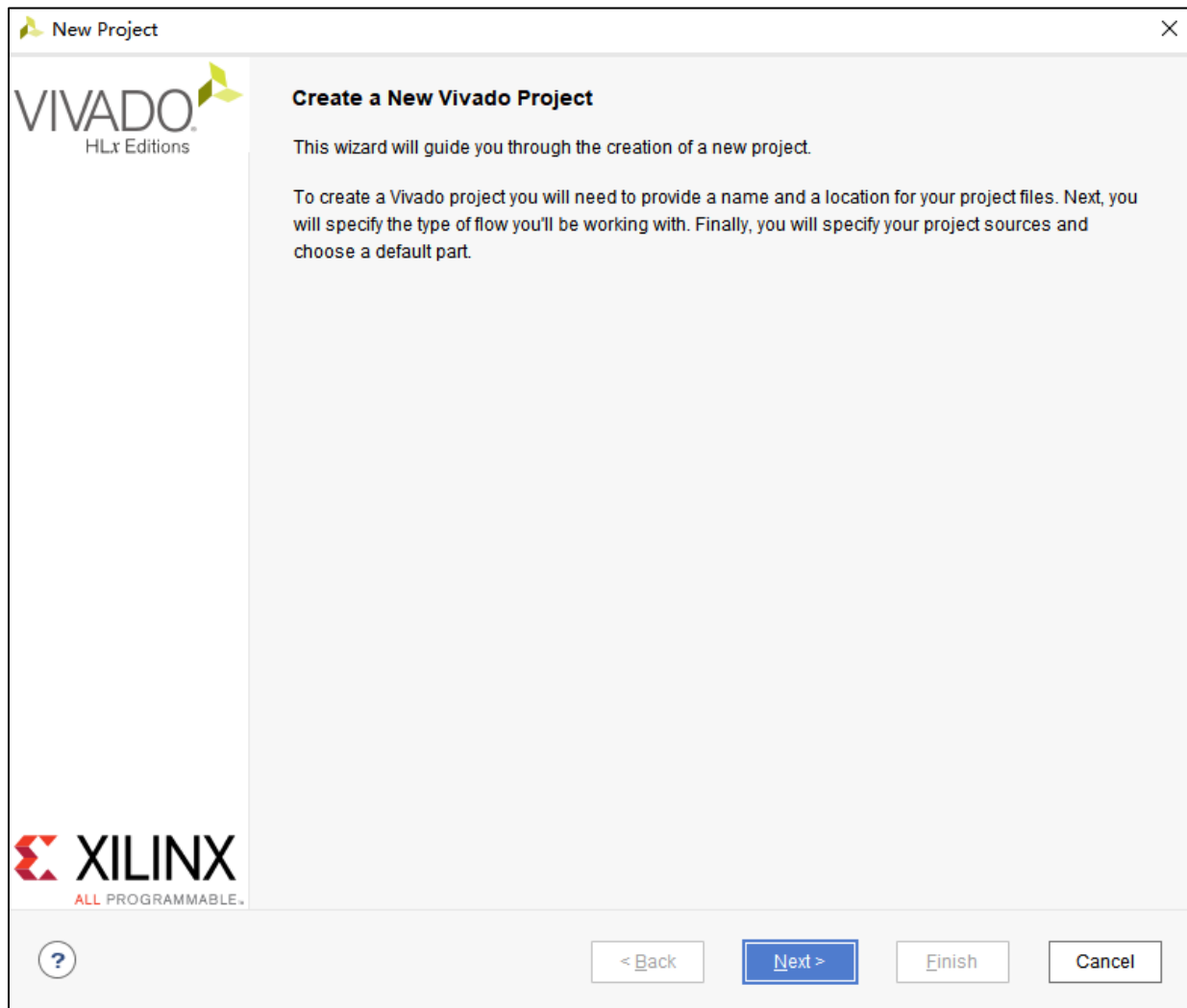
Learning Center

[Documentation and Tutorials >](#)
[Quick Take Videos >](#)
[Release Notes Guide >](#)

Vivado开发工具简介



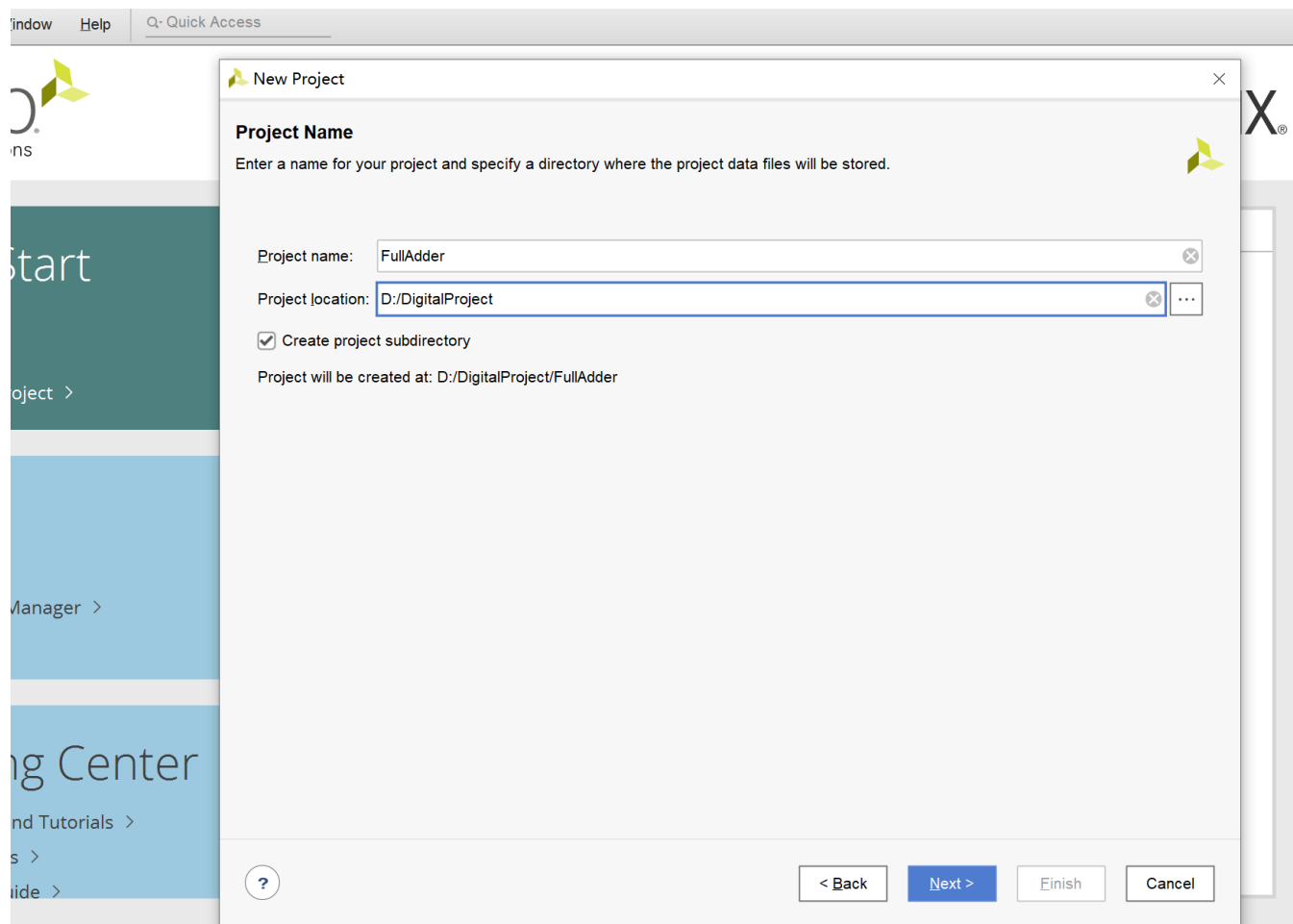
第二步：点Create New Project …进入新建工程向导



Vivado开发工具简介



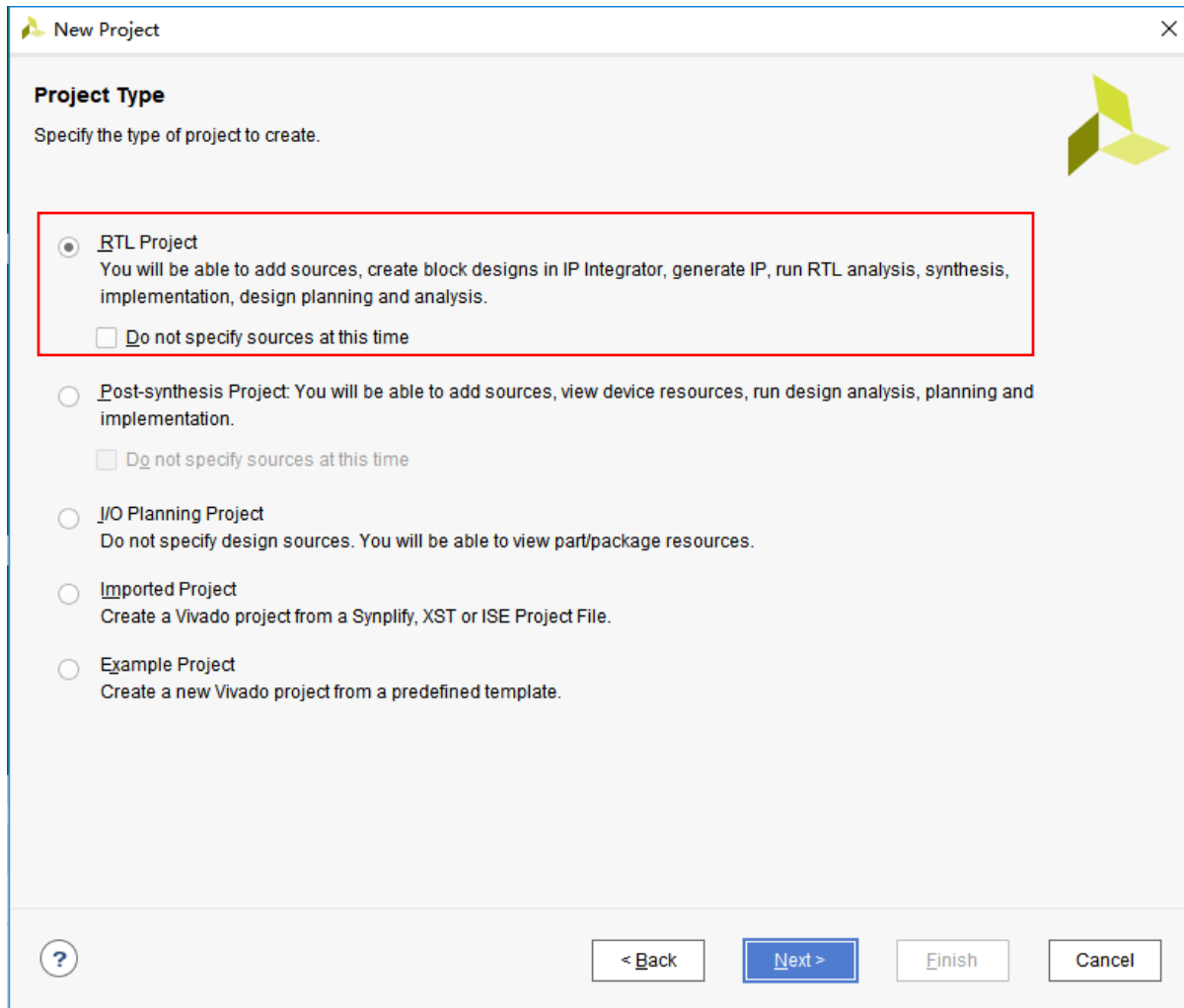
第三步： 在弹出窗口填写Project名称， 并选择存储地址



Vivado开发工具简介



在弹出窗口按下图设置：



Vivado开发工具简介



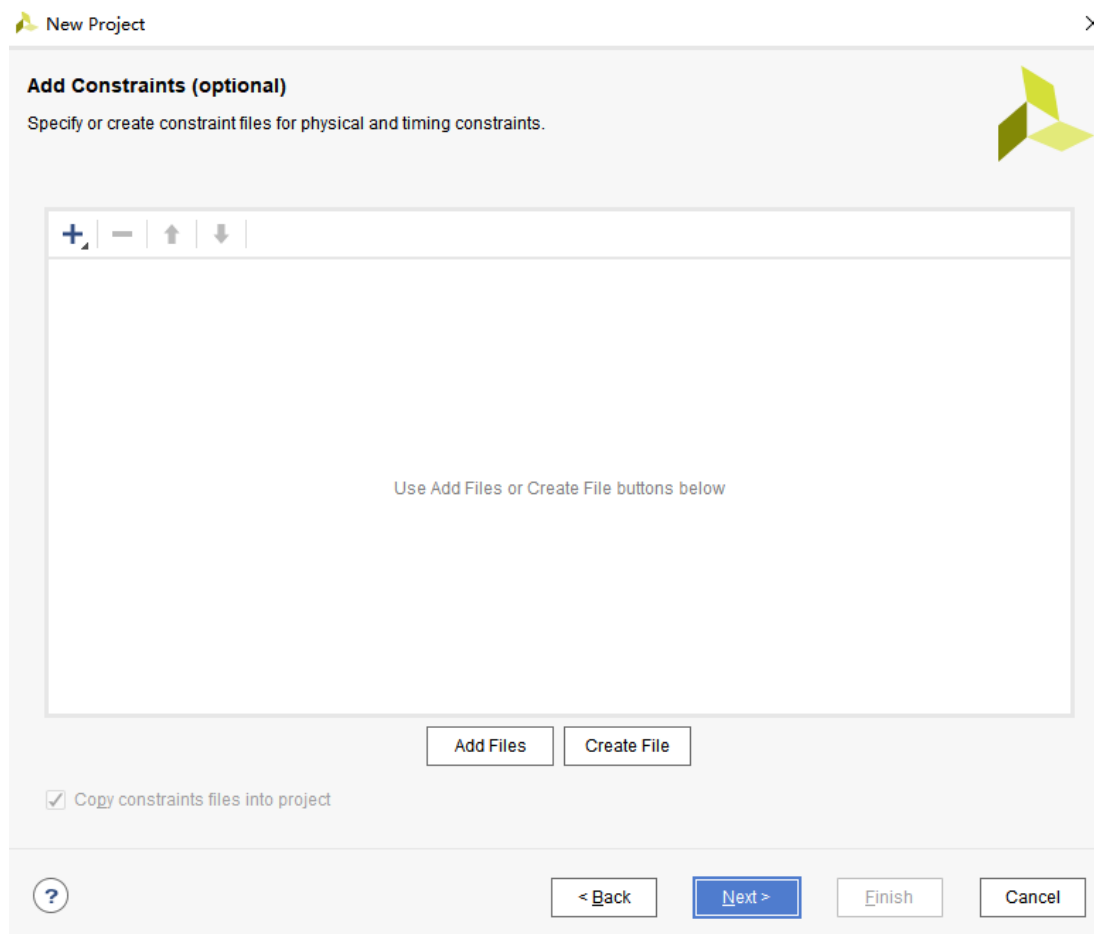
由于我们是新建工程，所以此处默认没有可以添加的源文件，并且设置编程语言和仿真语言均设置为VHDL。点击Next

The image shows the 'New Project' dialog box in Vivado, specifically the 'Add Sources' step. The window title is 'New Project'. The main heading is 'Add Sources'. Below this, a text block explains: 'Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.' To the right of this text is a small green icon. Below the text is a large, empty rectangular area with a toolbar at the top containing icons for adding (+), removing (-), moving up (↑), and moving down (↓). Inside this area, a message reads: 'Use Add Files, Add Directories or Create File buttons below'. Below this area are three buttons: 'Add Files', 'Add Directories', and 'Create File'. Further down are three checkboxes: 'Scan and add RTL include files into project' (unchecked), 'Copy sources into project' (checked), and 'Add sources from subdirectories' (checked). At the bottom of this section are two dropdown menus: 'Target language:' set to 'VHDL' and 'Simulator language:' set to 'VHDL'. The bottom of the dialog features a navigation bar with a help icon (?), '< Back', 'Next >' (highlighted in blue), 'Finish', and 'Cancel' buttons.

Vivado开发工具简介



也没有可以添加的约束文件，所以不添加，点击Next如下图：



Vivado开发工具简介



选择FPGA芯片，如下图选择xc7k160tffg676-2，并点击下一步

Default Part
Choose a default Xilinx part or board for your project.

Parts | Boards

[Reset All Filters](#)

Category: All Package: ffg676 Temperature: All Remaining
Family: Kintex-7 Speed: -2 Static power: All Remaining

Search: Q-

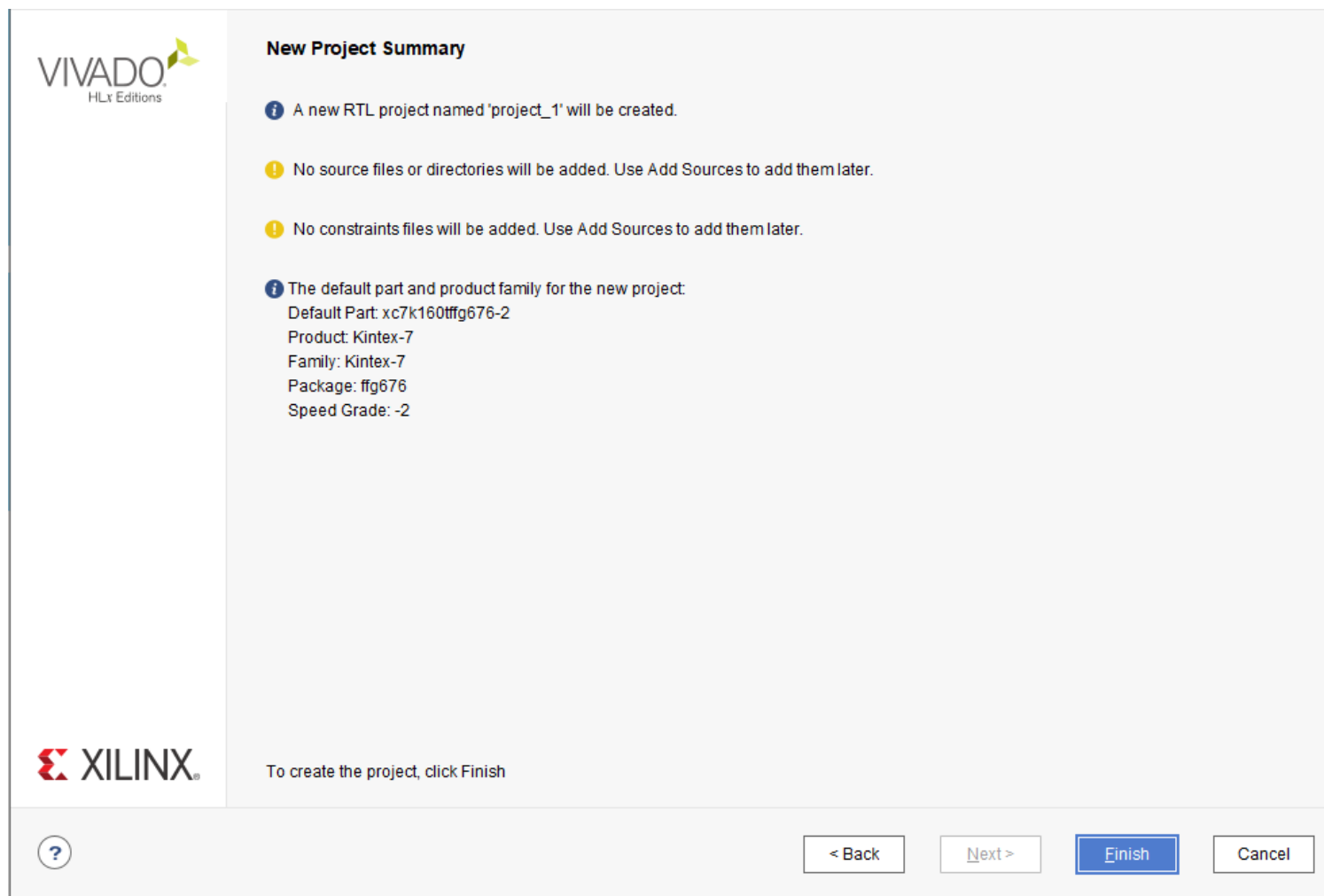
Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs	Gb Transceivers	GTPE2 T
xc7k160tffg676-2	676	400	101400	202800	325	0	600	8	0
xc7k325tffg676-2	676	400	203800	407600	445	0	840	8	0
xc7k410tffg676-2	676	400	254200	508400	795	0	1540	8	0

[?](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

Vivado开发工具简介



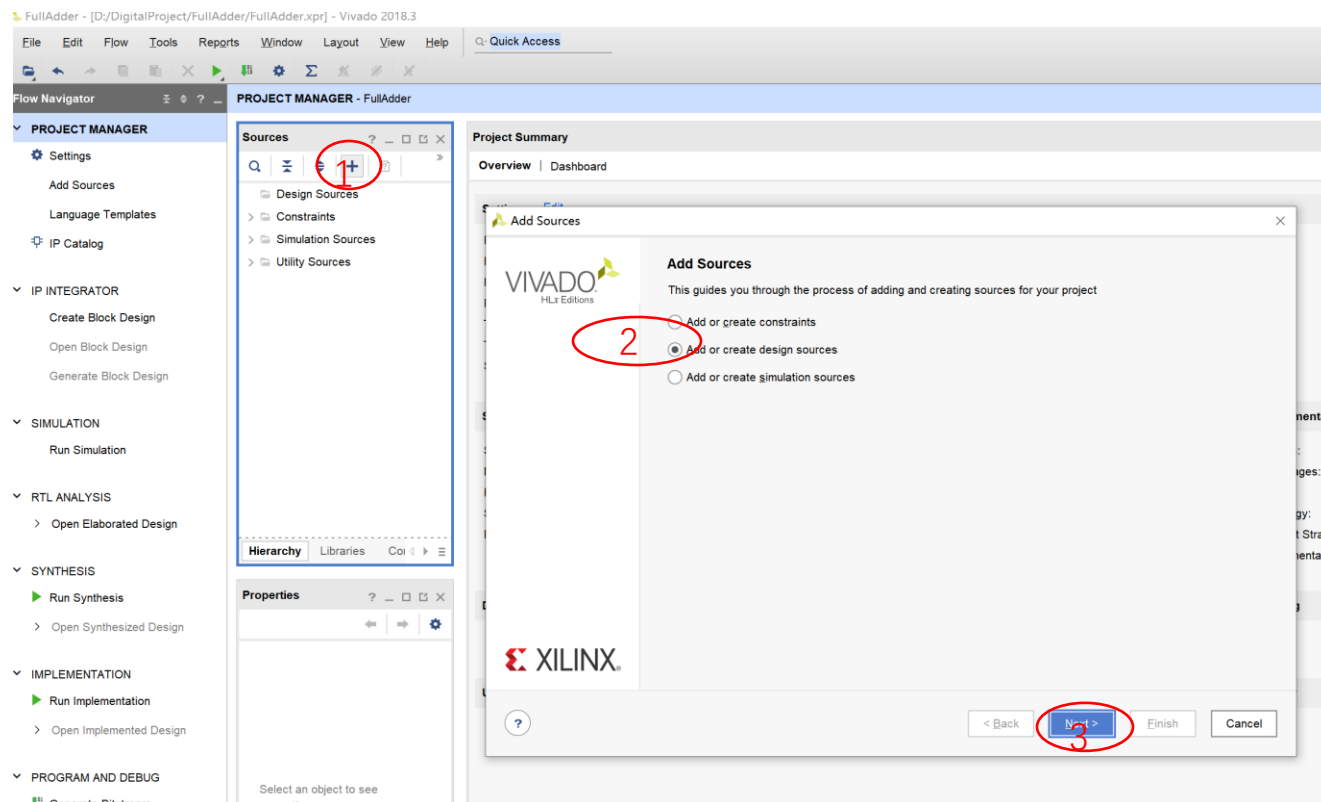
查看工程总况，点击Finish



Vivado开发工具简介



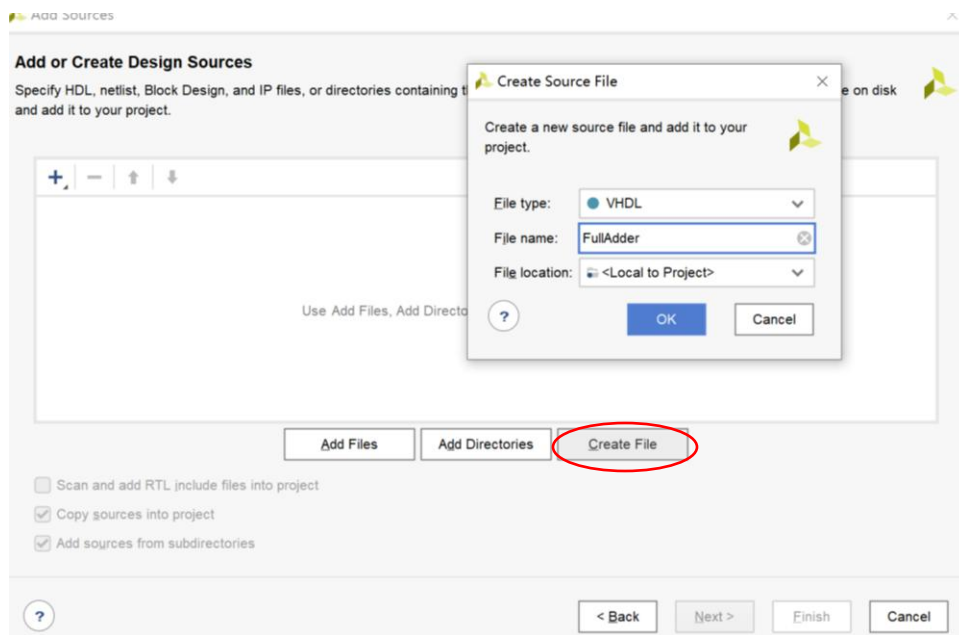
第四步：新建（添加）源文件



Vivado开发工具简介

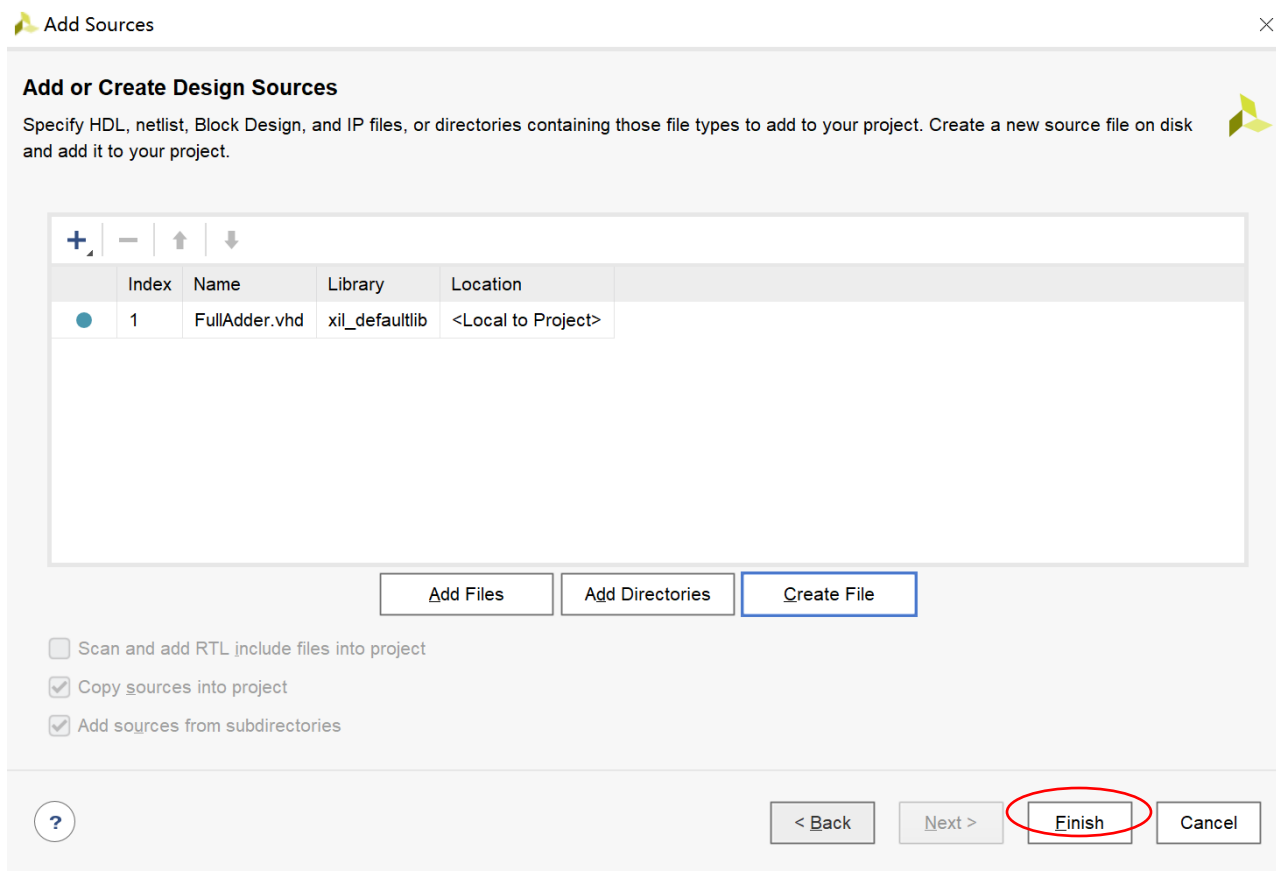


在上图中选择Create File…在弹出下面窗口填写新建源文件名称。



点击OK，弹出下图：

Vivado开发工具简介



在上图中点击Finish在弹出窗口填写模块名称和端口，如下图所示

Vivado开发工具简介



Define Module ×

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Entity name: ×

Architecture name: ×

I/O Port Definitions

+

-

↑

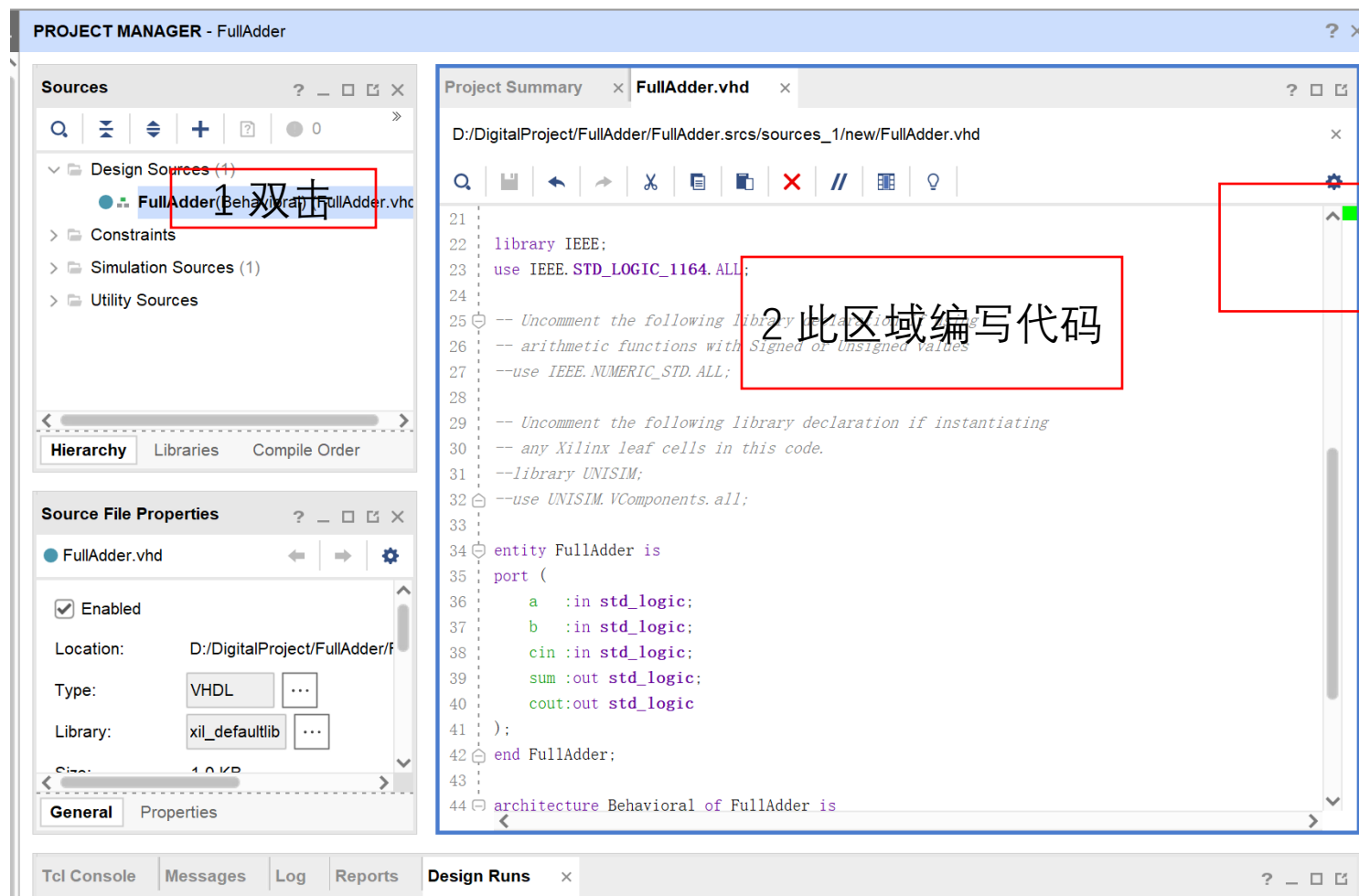
↓

Port N...	Dire...	...	...	...
	in	▼	☐	0 0

? OK Cancel

端口名称一般不写，在后面写代码的时候再定义，点击ok，弹出窗口继续点YES

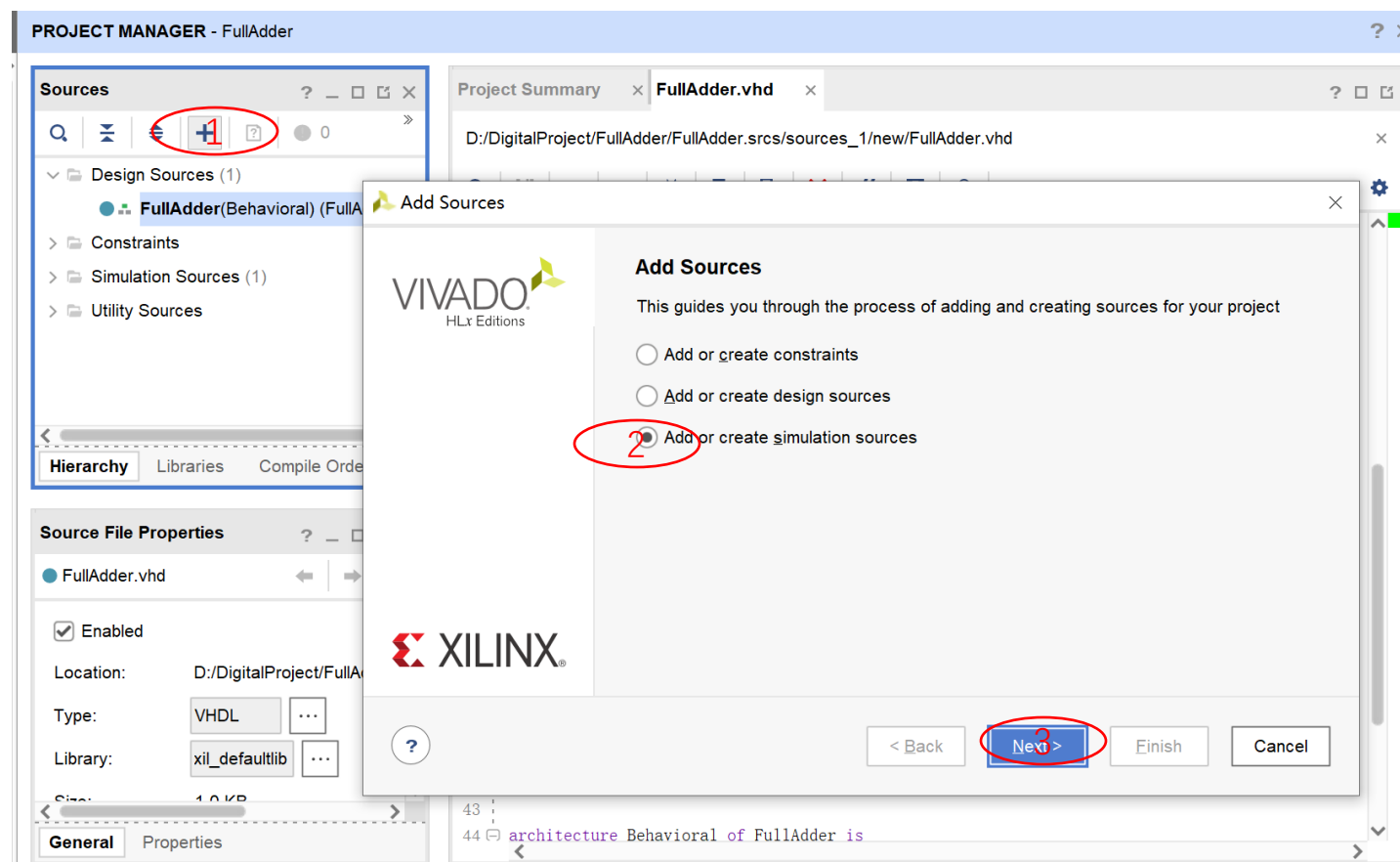
Vivado开发工具简介



Vivado开发工具简介



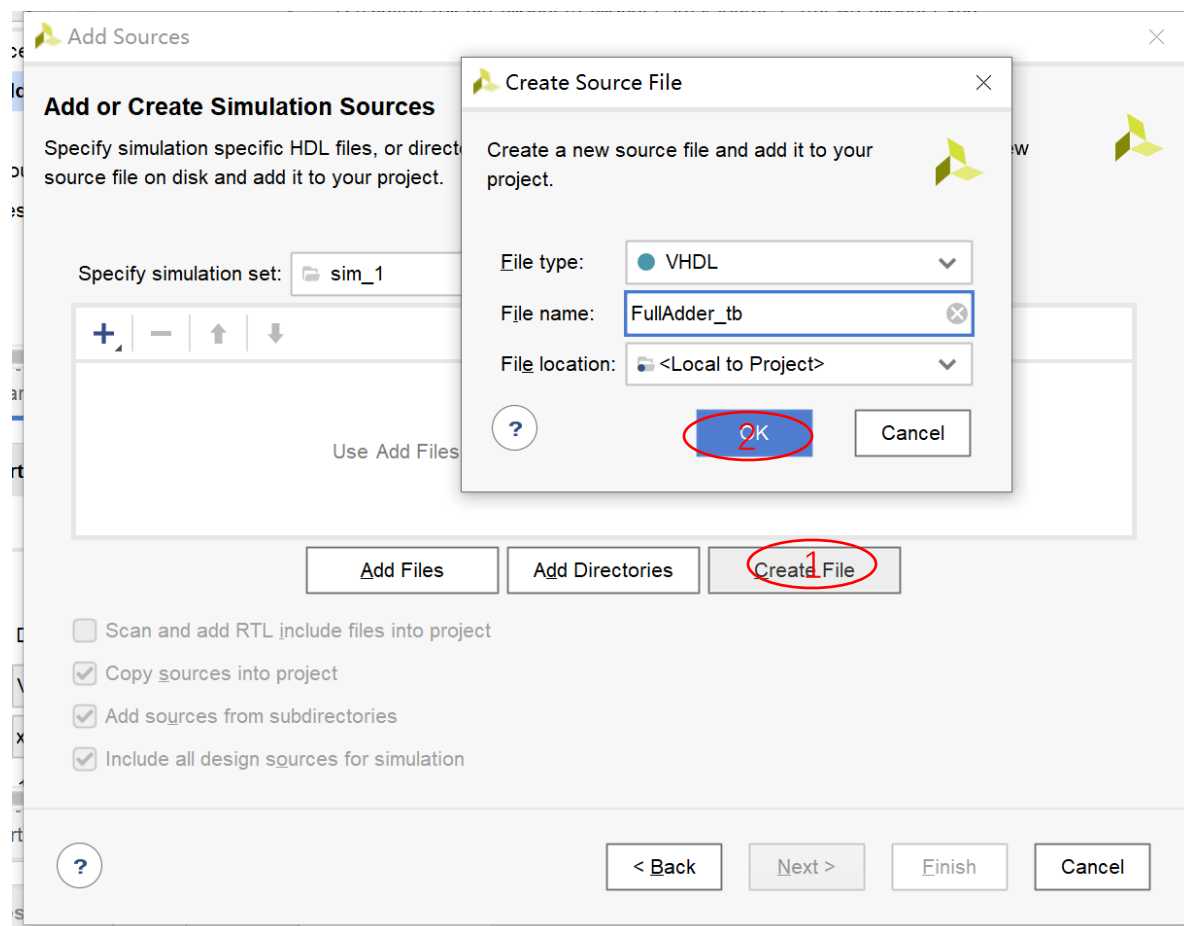
第五步：点击加号，选择添加仿真文件



Vivado开发工具简介



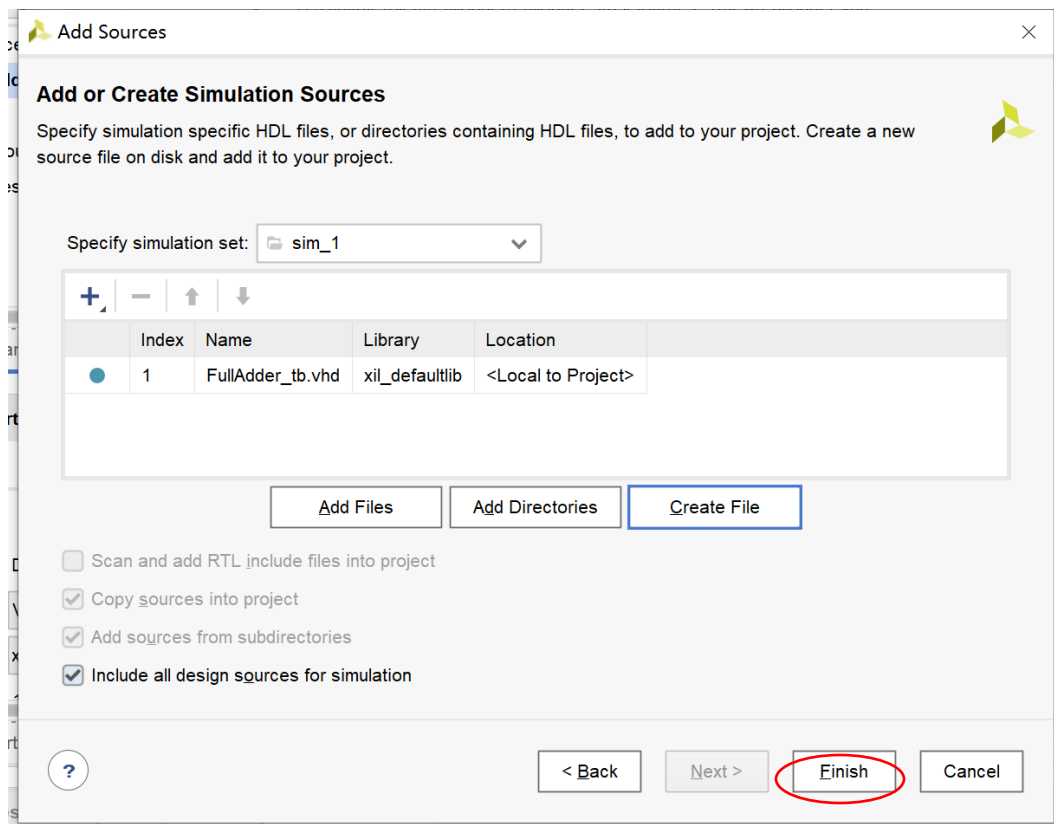
在下图的弹出窗口选择Create File...



Vivado开发工具简介



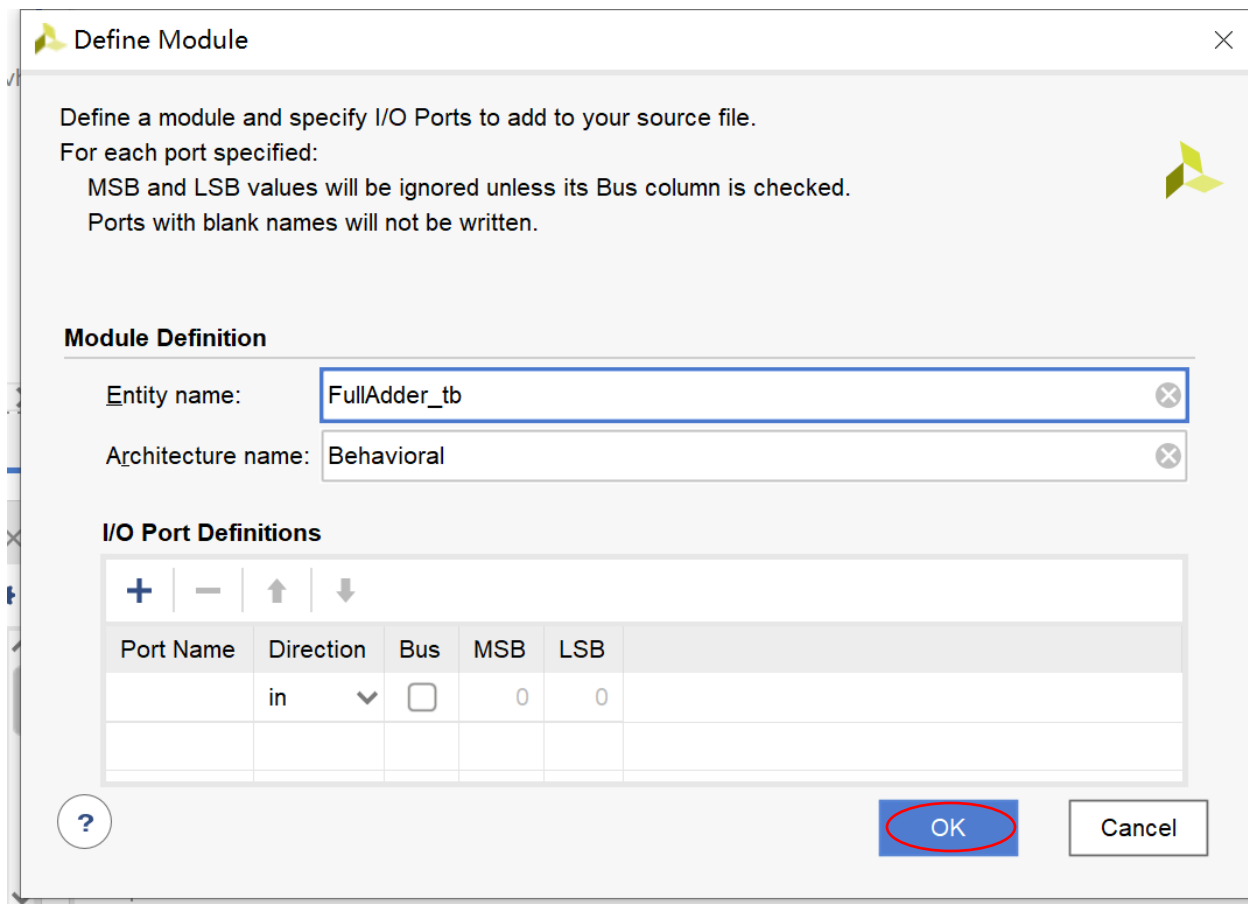
点击Finish



Vivado开发工具简介



点击OK，弹出窗口后然后点YES



Define Module

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Entity name: FullAdder_tb

Architecture name: Behavioral

I/O Port Definitions

Port Name	Direction	Bus	MSB	LSB
	in	☐	0	0

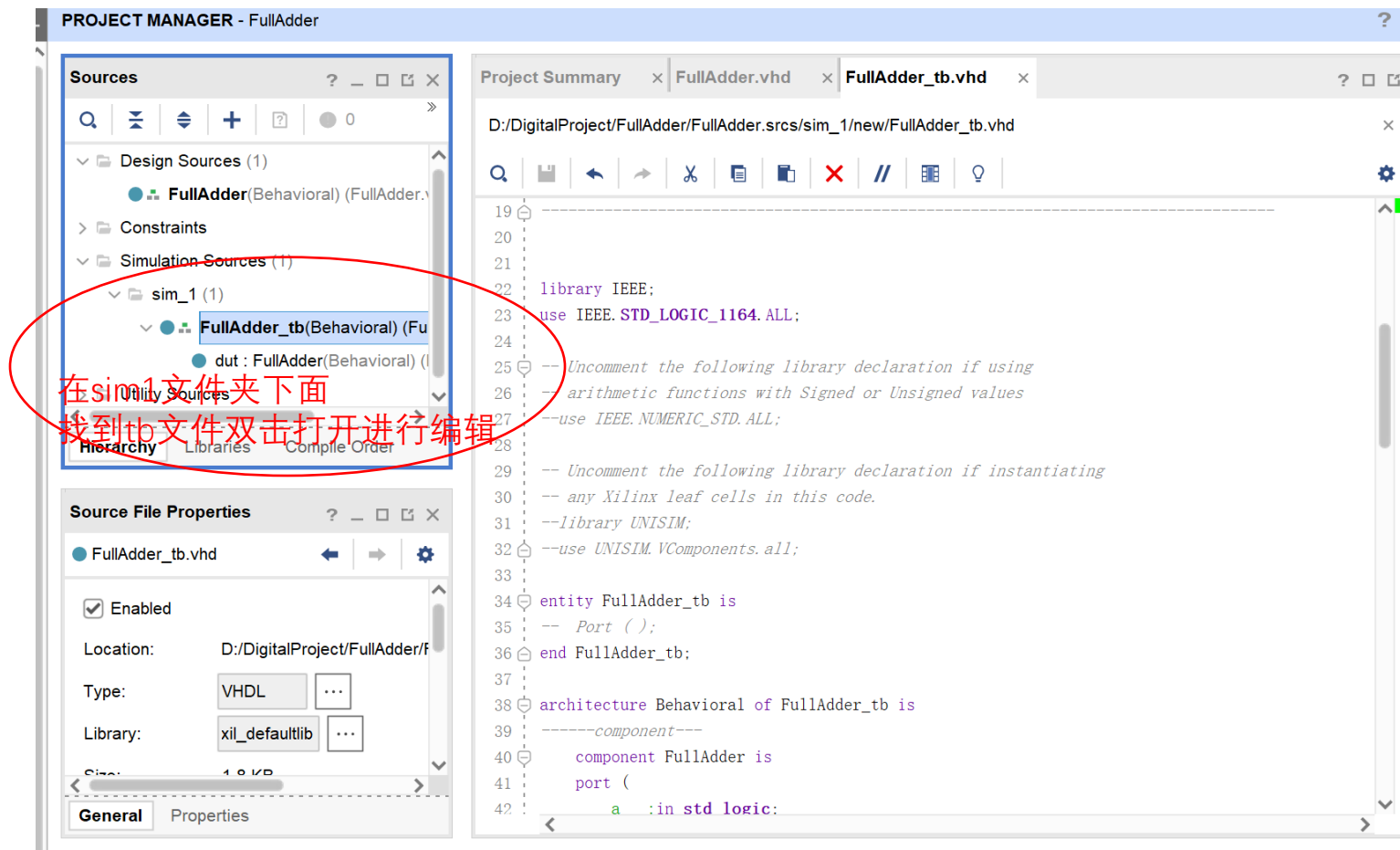
?

OK Cancel

Vivado开发工具简介



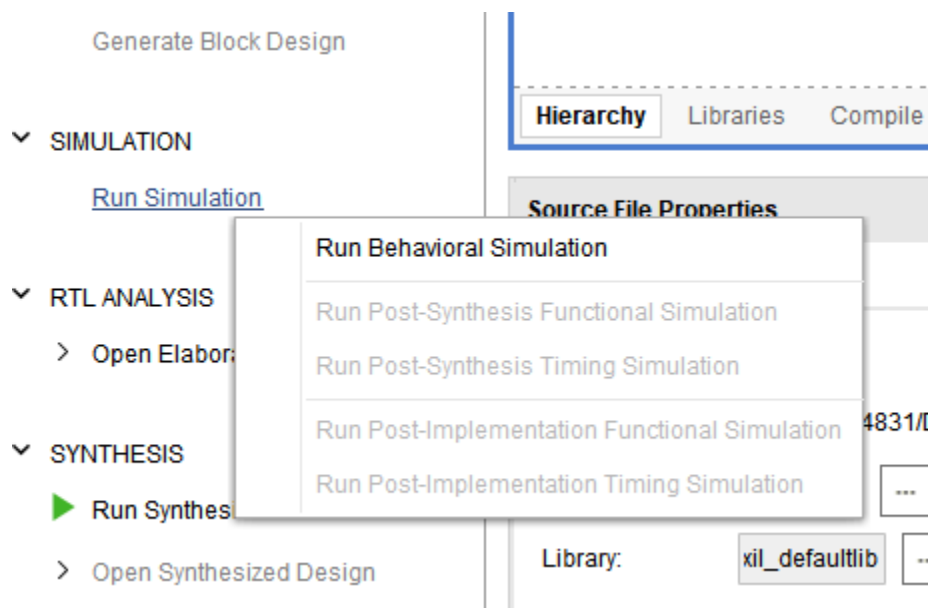
双击打开并编写testbench文件



Vivado开发工具简介



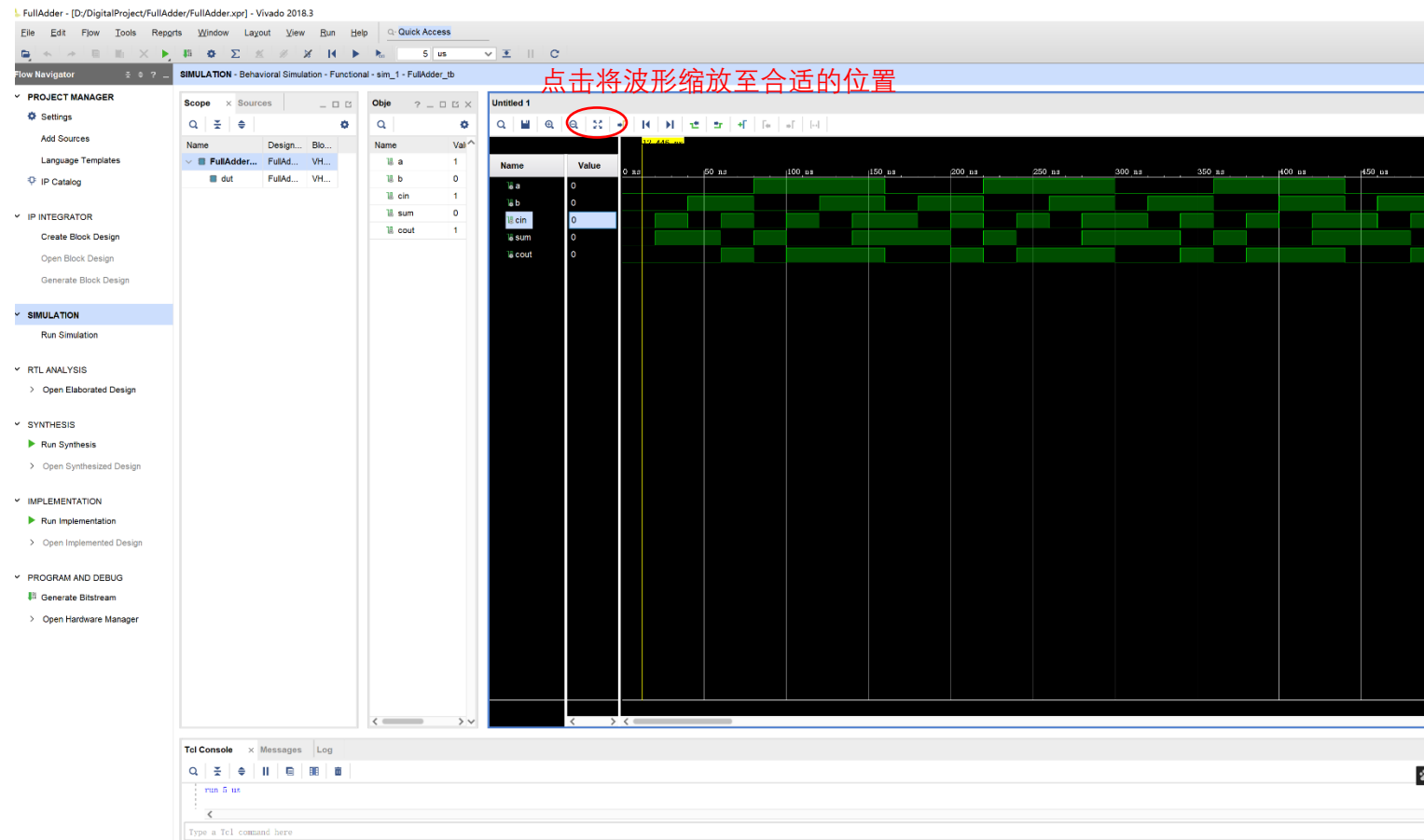
编写完仿真源文件后，单击Run Simulation，选择Run Behavior Simulation运行仿真



Vivado开发工具简介



弹出如下的窗口



全加器仿真例程



FullAdder_tb.vhd

FullAdder.vhd

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity FullAdder is
5  port (
6      a      : in std_logic;
7      b      : in std_logic;
8      cin    : in std_logic;
9      sum    : out std_logic;
10     cout   : out std_logic
11 );
12 end entity;
13
14 architecture Behavioral of FullAdder is
15
16     begin
17         sum    <= a xor b xor cin;
18         cout   <= (a and b) or ((a xor b) and cin);
19     end architecture;
```

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity FullAdder_tb is
5  end entity;
6
7  architecture Behavioral of FullAdder_tb is
8  component FullAdder is
9  port (
10      a      : in std_logic;
11      b      : in std_logic;
12      cin    : in std_logic;
13      sum    : out std_logic;
14      cout   : out std_logic
15  );
16  end component;
17
18  signal a      : std_logic;
19  signal b      : std_logic;
20  signal cin    : std_logic;
21  signal sum    : std_logic;
22  signal cout   : std_logic;
23
24  begin
25      dut:FullAdder port map(
26          a      => a ,
27          b      => b ,
28          cin    => cin ,
29          sum    => sum ,
30          cout   => cout
31      );
32
33  process begin
34      a<='0'; b<='0'; cin<='0'; wait for 20 ns;
35      a<='0'; b<='0'; cin<='1'; wait for 20 ns;
36      a<='0'; b<='1'; cin<='0'; wait for 20 ns;
37      a<='0'; b<='1'; cin<='1'; wait for 20 ns;
38      a<='1'; b<='0'; cin<='0'; wait for 20 ns;
39      a<='1'; b<='0'; cin<='1'; wait for 20 ns;
40      a<='1'; b<='1'; cin<='0'; wait for 20 ns;
41      a<='1'; b<='1'; cin<='1'; wait for 20 ns;
42  end process;
43
44  end architecture;
```



https://www.cs.sfu.ca/~ggbaker/reference/std_logic/

The `std_logic` Libraries

The [IEEE](#) created the IEEE VHDL library and `std_logic` type in standard 1164. This was extended by [Synopsys](#); their extensions are freely redistributable.

Parts of the IEEE library can be included in an entity by inserting lines like these before your entity declaration:

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
```

No attempt has been made here to be definitive or exhaustive. If you want real answers, read the source code. Links to the code or instructions on how to get it are provided for each section. The source is fairly readable to someone who knows some VHDL.

Library information

You can find information about the following libraries here:

- [std_logic_1164](#)
- [std_logic_arith](#)
- [std_logic_unsigned](#)
- [std_logic_signed](#)

Missing parts

Some more of the libraries will be added here eventually (hopefully). Until then, I will provide some source code:

- [std_logic_entities](#)
- [std_logic_components](#)
- [std_logic_misc](#)
- [std_logic_textio](#)
- [qs_types](#)
- [cyclone_resolved](#)

[About these pages](#)

Return to [Greg's References](#).

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实验任务



- 完成一个4比特加法器 $S = A + B$
- 输入：A,B均为4比特二进制补码表征
- 输出：S的位宽自己定(计算溢出？饱和截位？)
- 加法器结构自己定：行波进位，超前进位，进位选择等任意加法器结构
- 自行设计testbench完成RTL行为级仿真
- 验收：以小组为单位完成

